Spin-Orbit Torque and Nernst Effects in Bi_xSb_{1-x}/Ferromagnet Heterostructures

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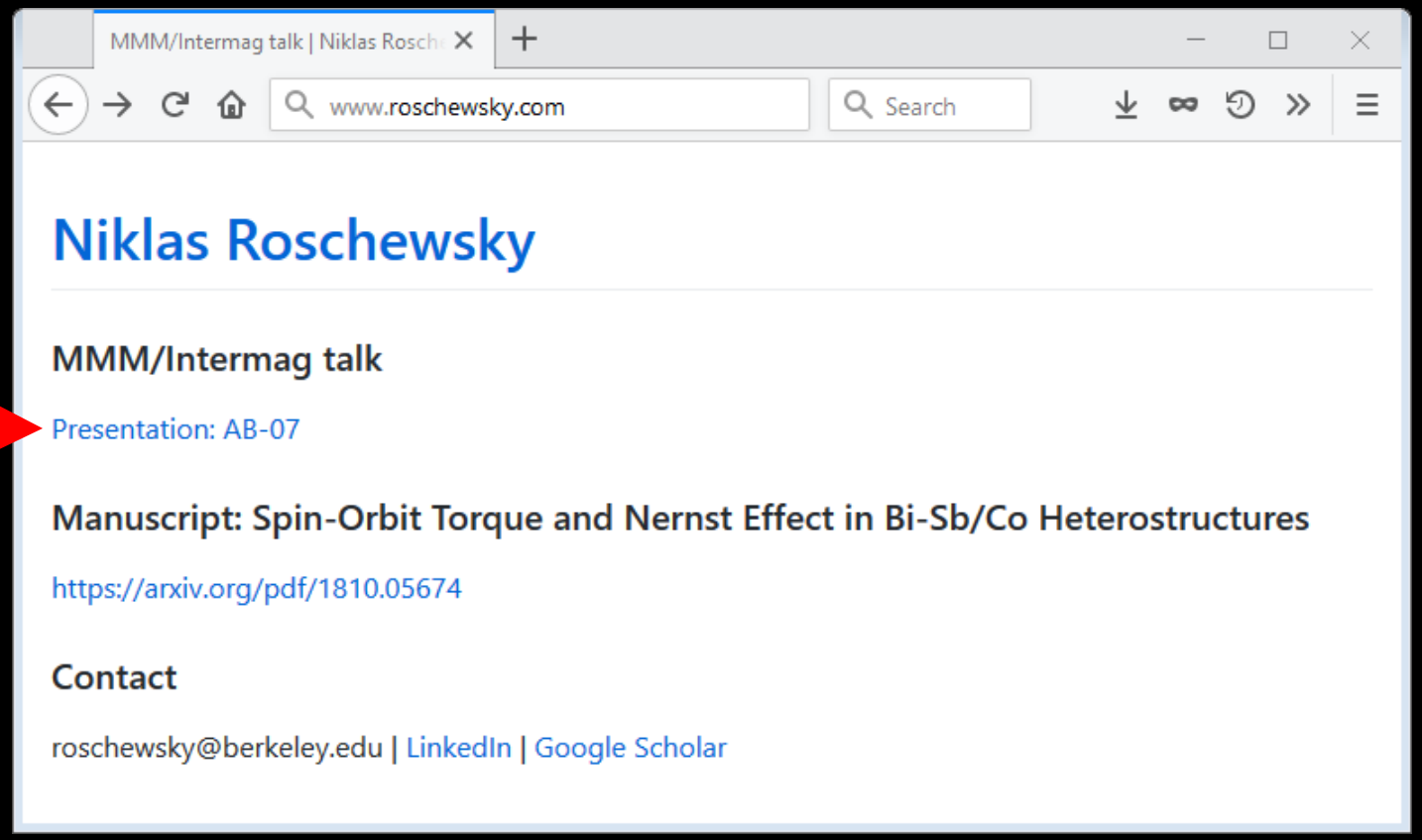
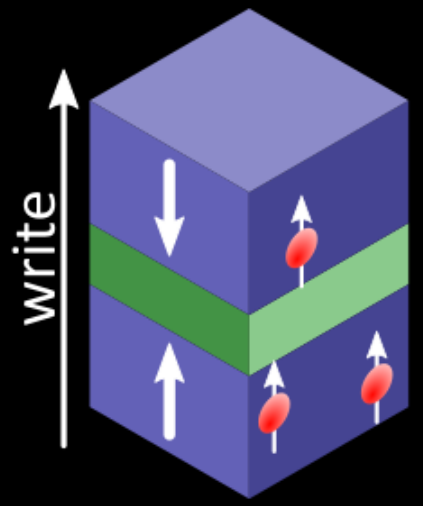


Spin-Torque Memory

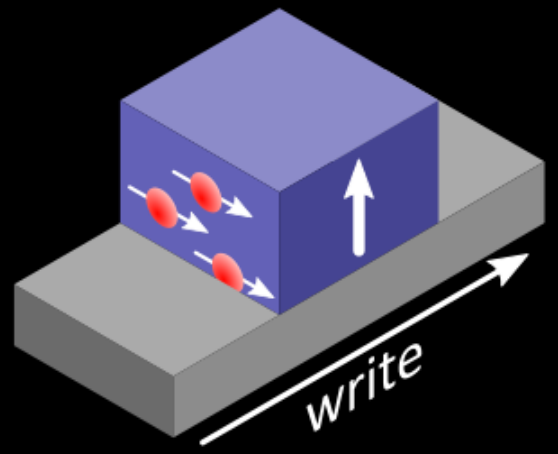
Spin-Transfer Torque - MRAM



- Fast
- Reliable
- High efficiency j_c/Δ

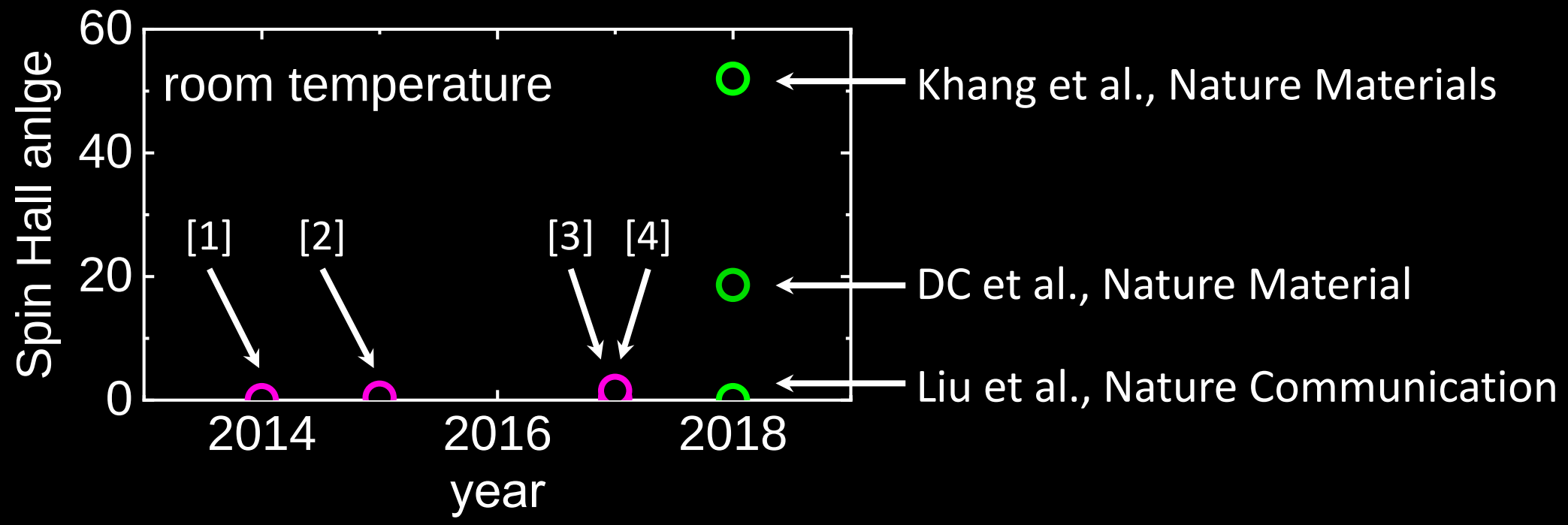
Spin-Orbit Torque - MRAM

- Faster
- Better Reliability
- Low efficiency j_c/Δ



Is low SOT efficiency j_c/Δ a materials challenge?

Spin Hall Angle in Topological Insulators

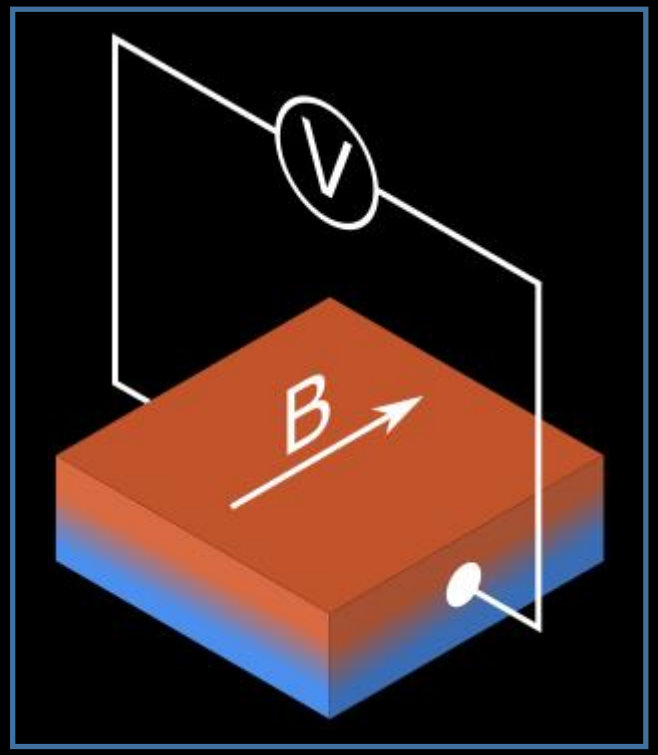


[1] Deorani et al., Physical Review B
[2] Jamali et al., Nano Letters
[3] Han et al., Physical Review Letters
[4] Wang et al., Nature Communications

Ordinary Nernst Effect in Topological Insulators

The ordinary Nernst effect is large in topological insulators

This might lead to an overestimation of the Spin Hall Angle



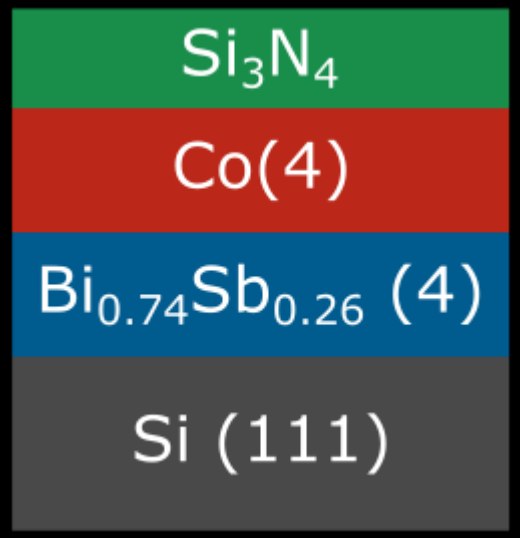
Harmonic Hall voltage persists in control samples without magnet

Harmonic Hall voltage is not suppressed at large magnetic field

A longitudinal electric field of same magnitude is generated

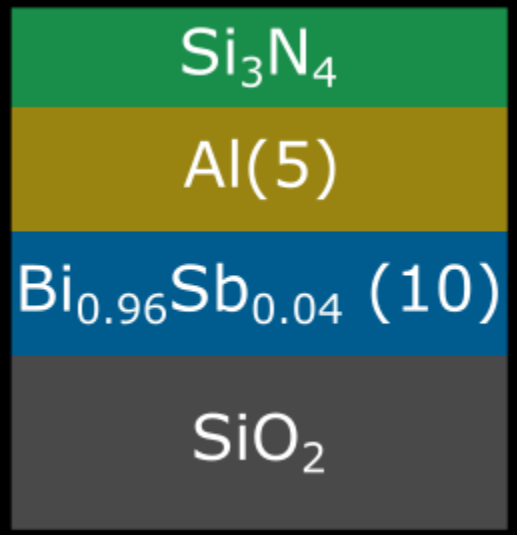
Sample Overview

Bi-Sb/Co



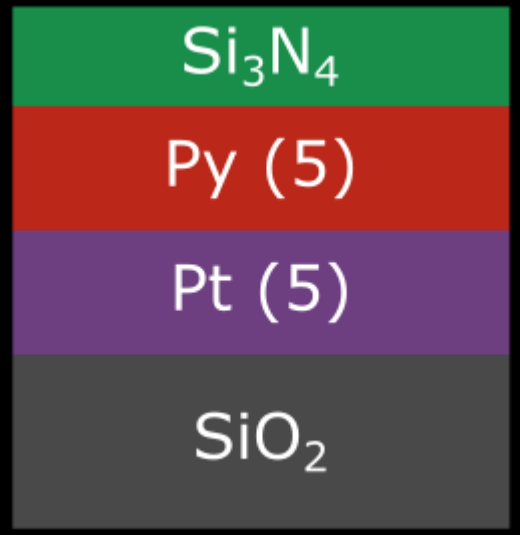
- Spin-Orbit Torque
- Thermal Effects

Bi-Sb/Al



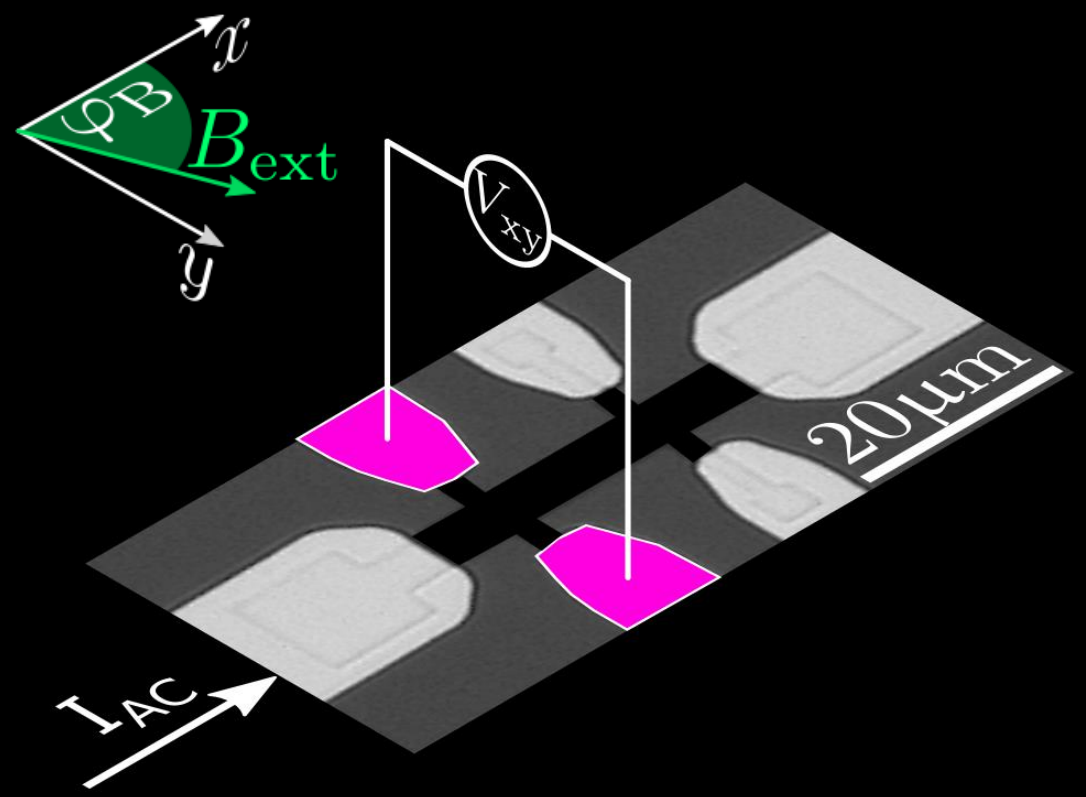
- No Spin-Orbit Torque
- Thermal Effects

Pt/Py



- Spin-Orbit Torque
- Small thermal effects

Harmonic Hall Measurements of SOT



Hall Voltage V_{xy}

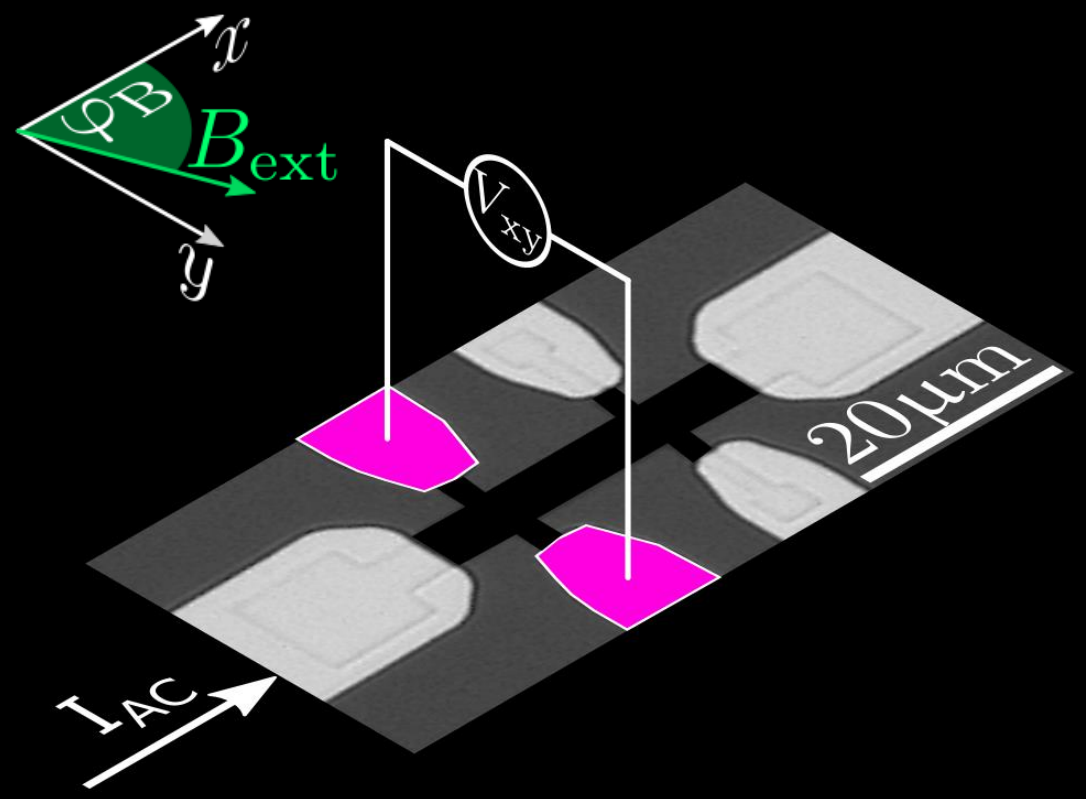
First Harmonic Hall Voltage

- Planar Hall Effect

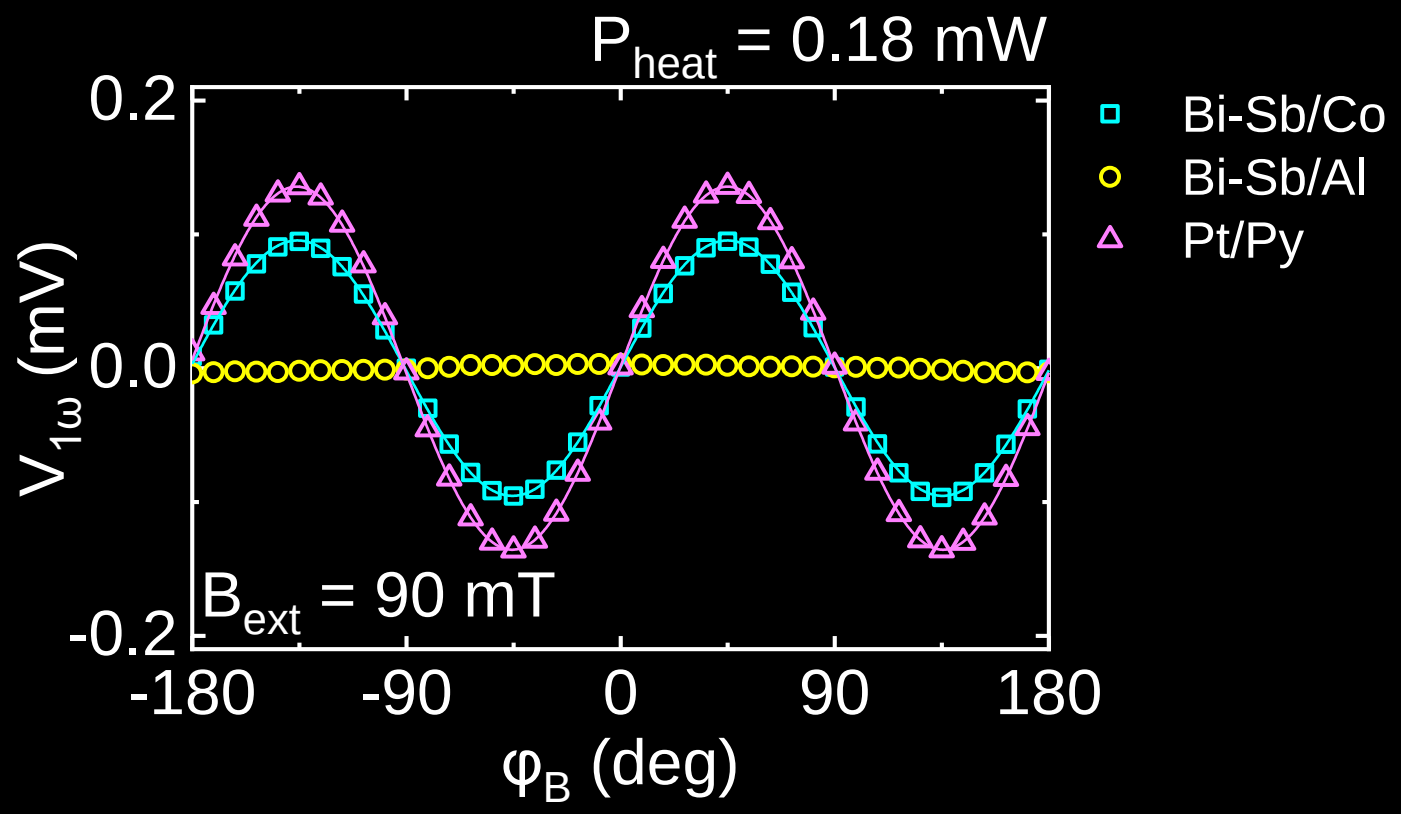
Second Harmonic Hall Voltage

- Spin orbit torque
- Thermal Effects

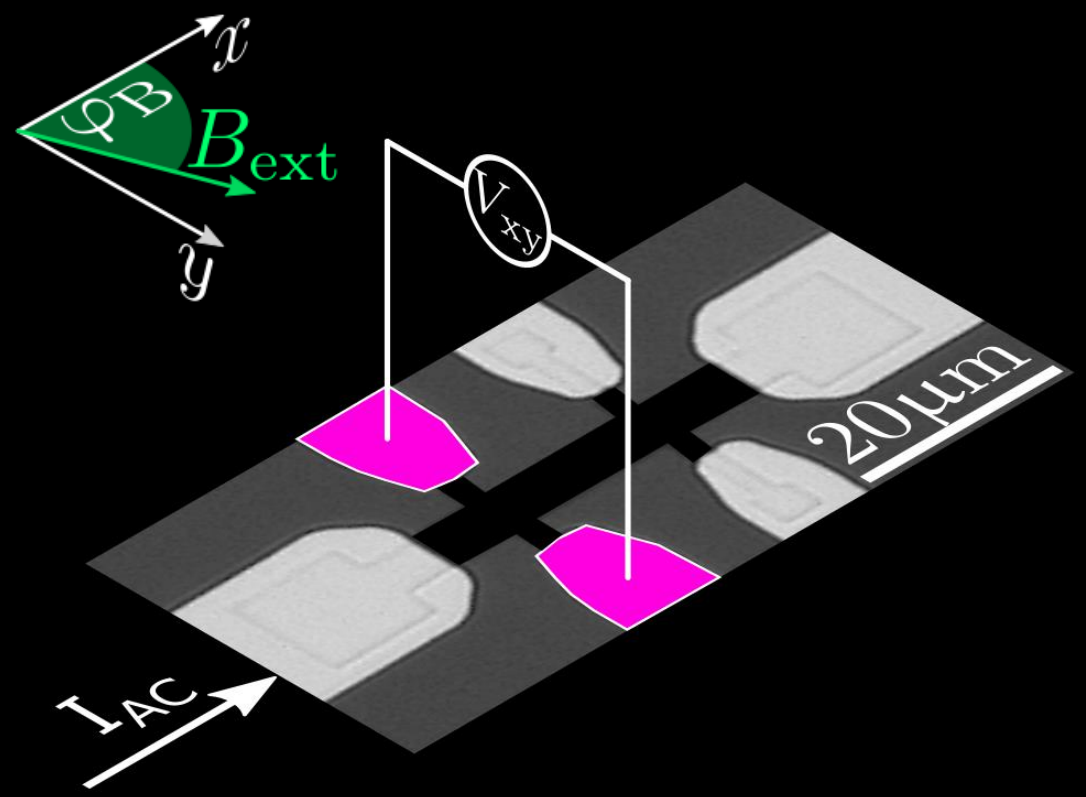
Harmonic Hall Measurements of SOT



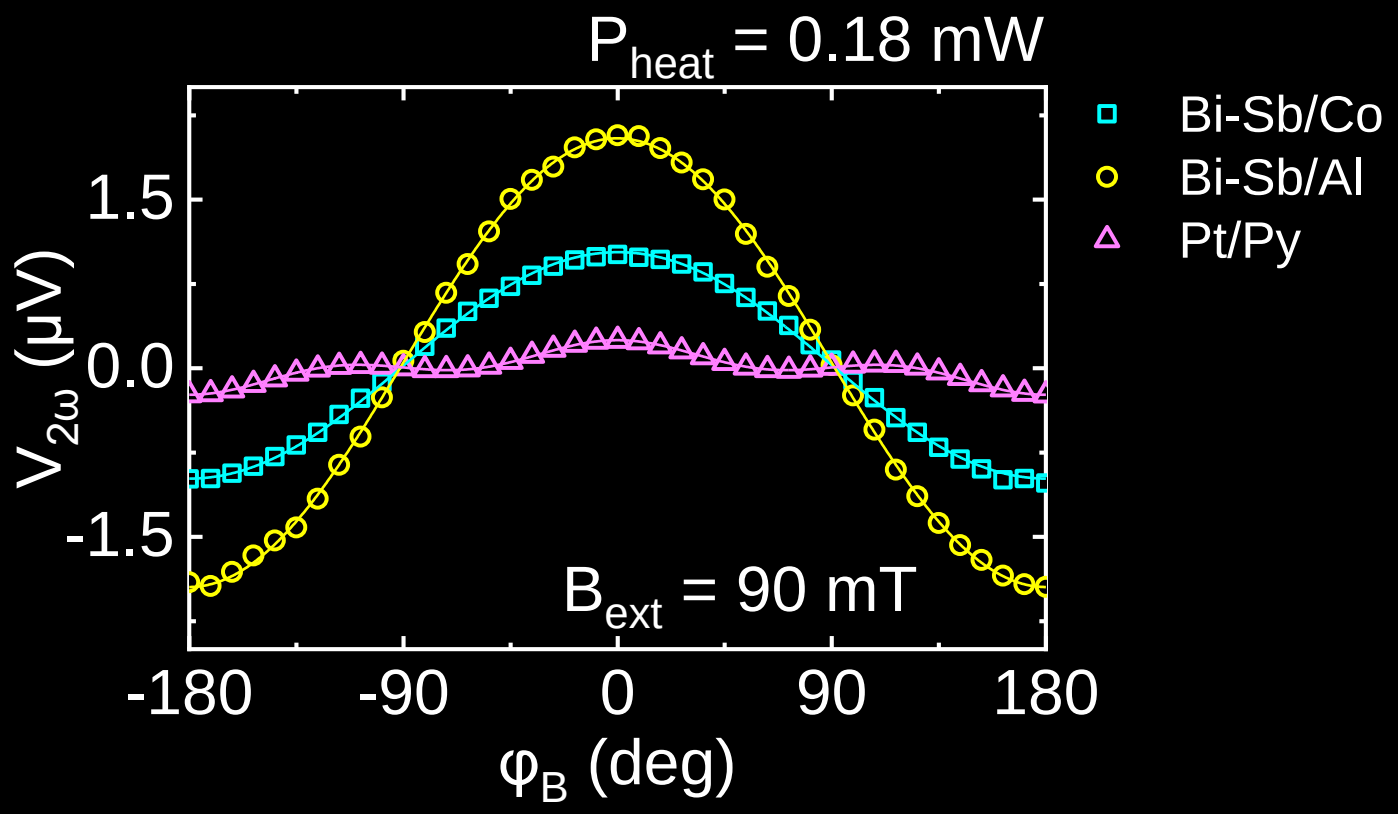
Hall Voltage V_{xy}



Harmonic Hall Measurements of SOT



Hall Voltage V_{xy}

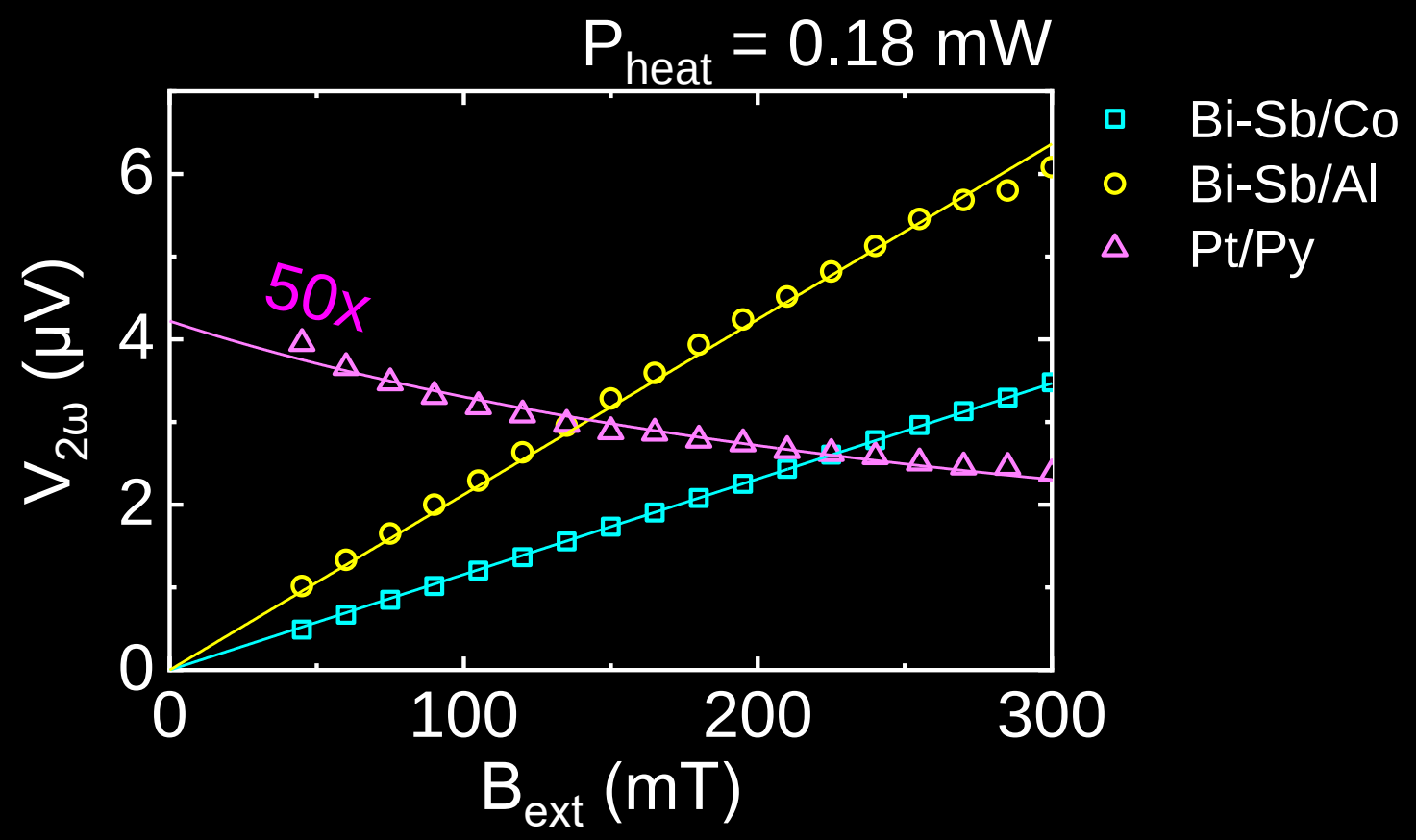


Harmonic Hall voltage persists in control samples without magnet

Harmonic Hall voltage is not suppressed at large magnetic field

A longitudinal electric field of same magnitude is generated

Magnetic Field Scaling



Linear scaling confirms presence of ordinary Nernst effect

Harmonic Hall voltage persists in control samples without magnet

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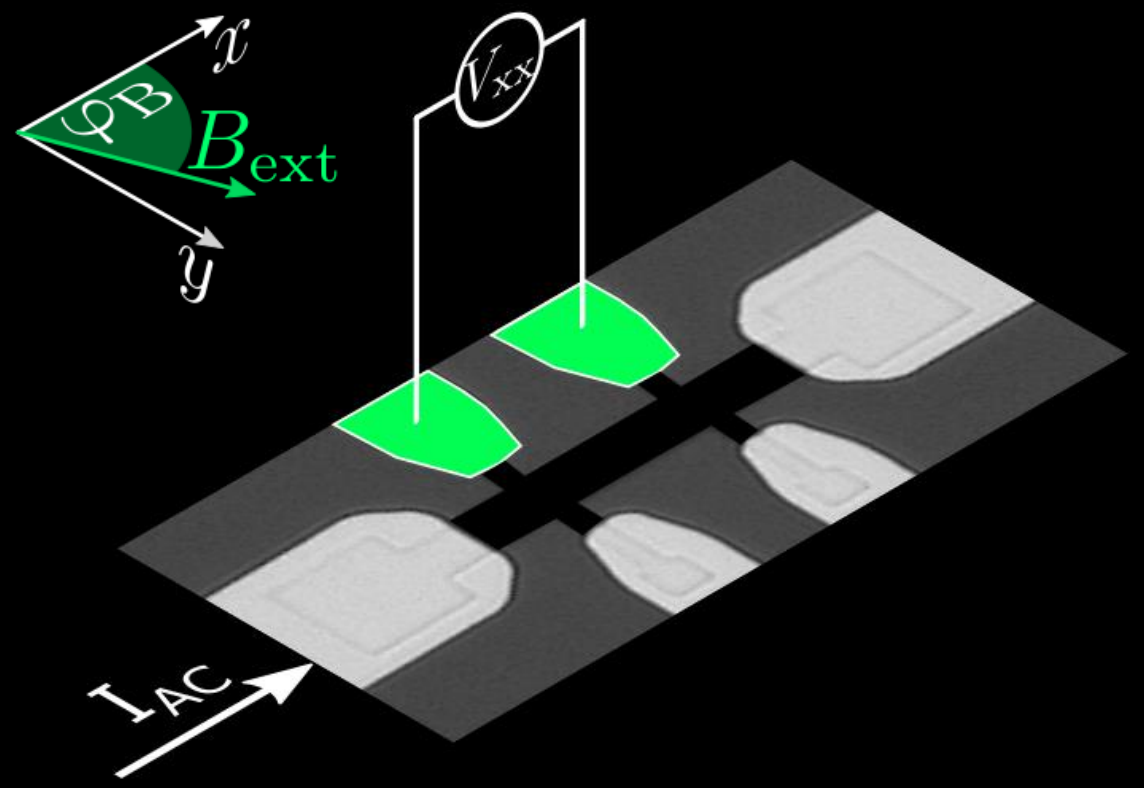
Longitudinal Voltage



First Harmonic Hall Voltage

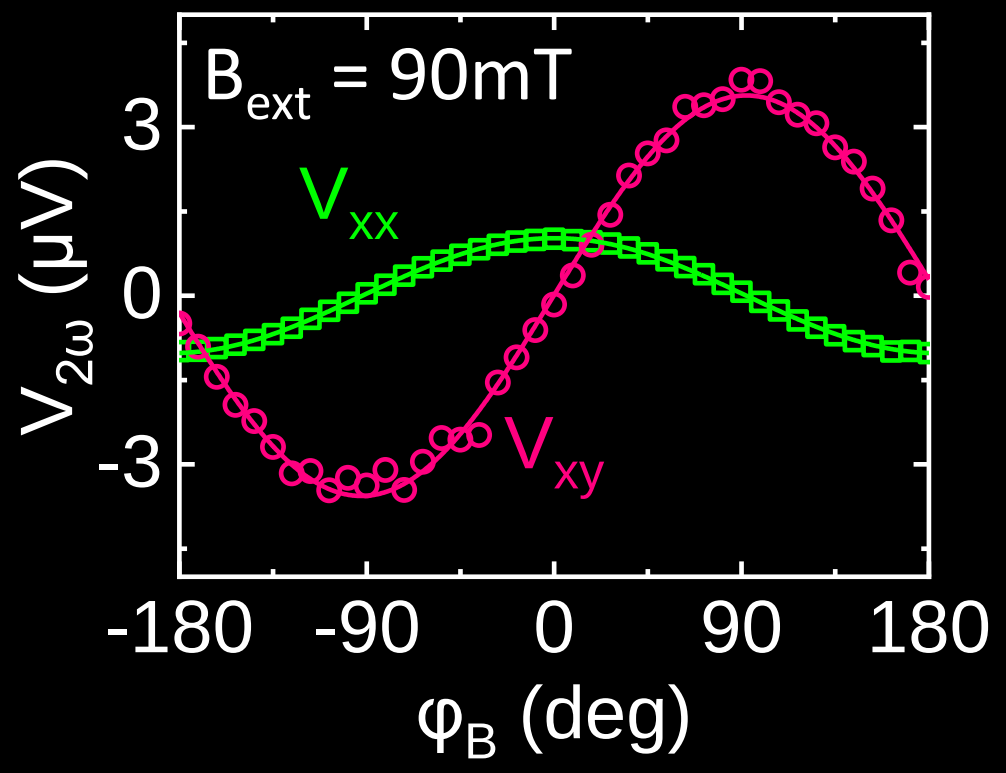
- Anisotropic Magnetoresistance

Second Harmonic Hall Voltage

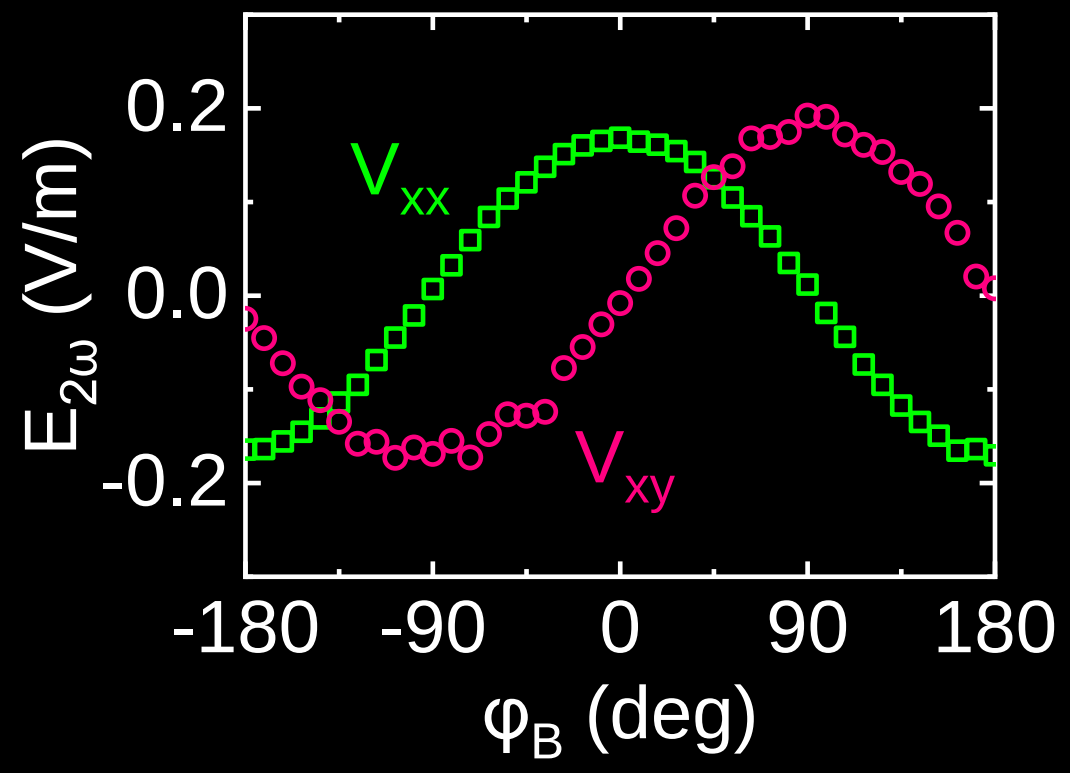
- Unidirectional Spin Hall MR
- Thermal Effects
- No damping-like SOT

Nernst Effect in Bi-Sb/Co

Voltage

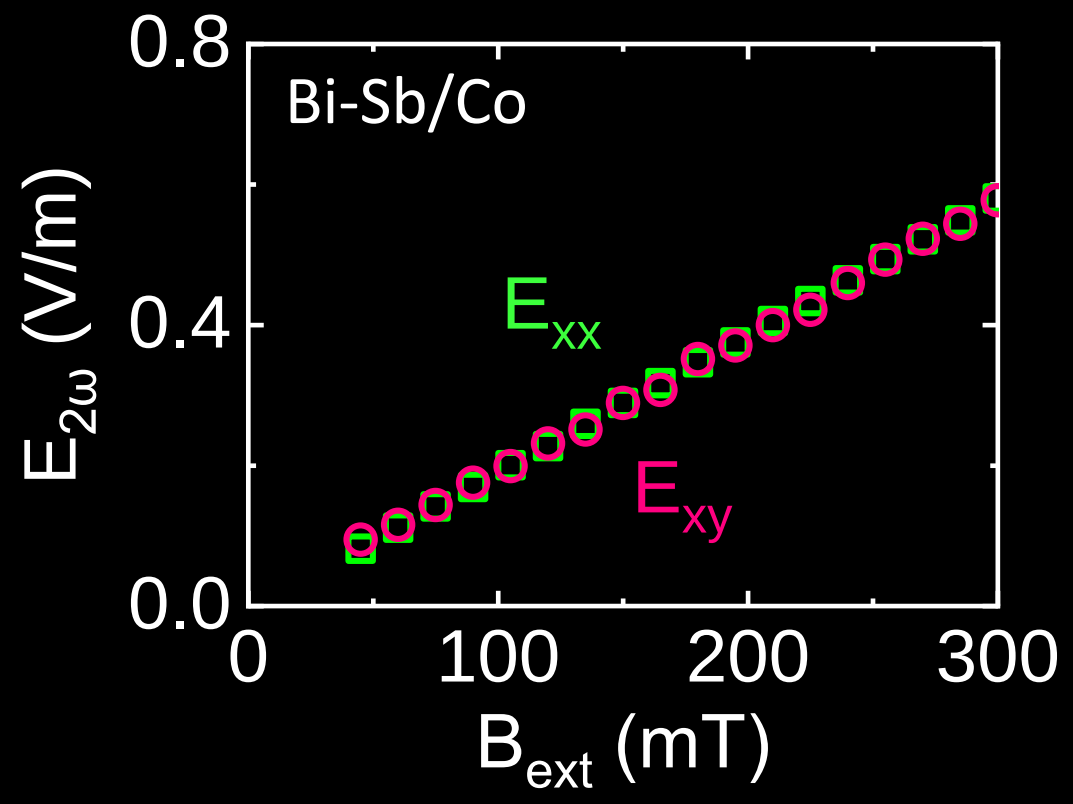


Electric Field

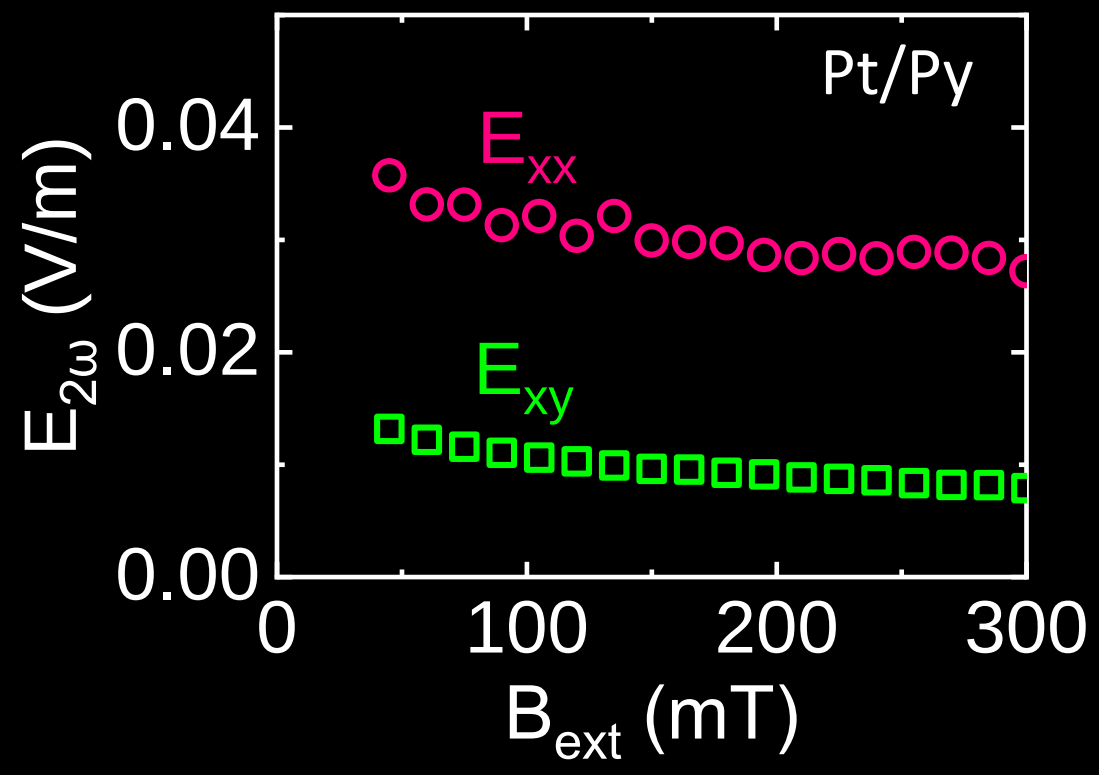


Spin Orbit Torque - Electric field

Ordinary Nernst Effect

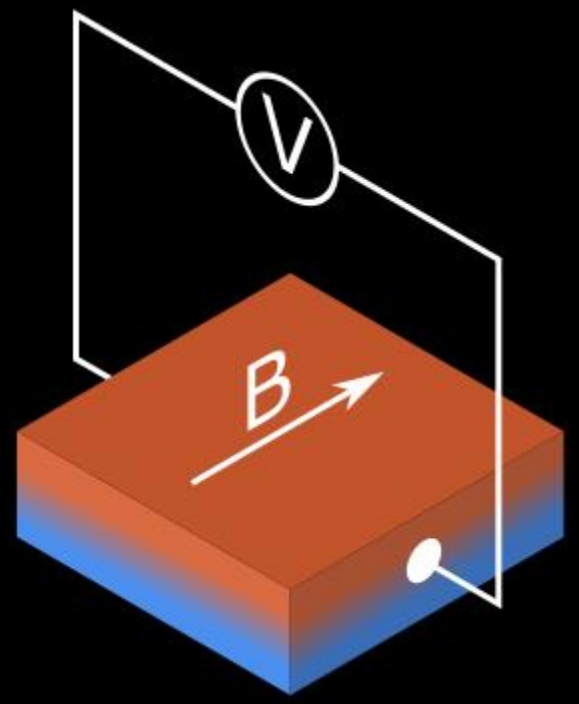


Spin-Orbit Torque & USMR



Conclusion

1. Nernst Effect can be large in semiconducting samples
2. Topological Insulators might not be an ideal material for MRAM applications



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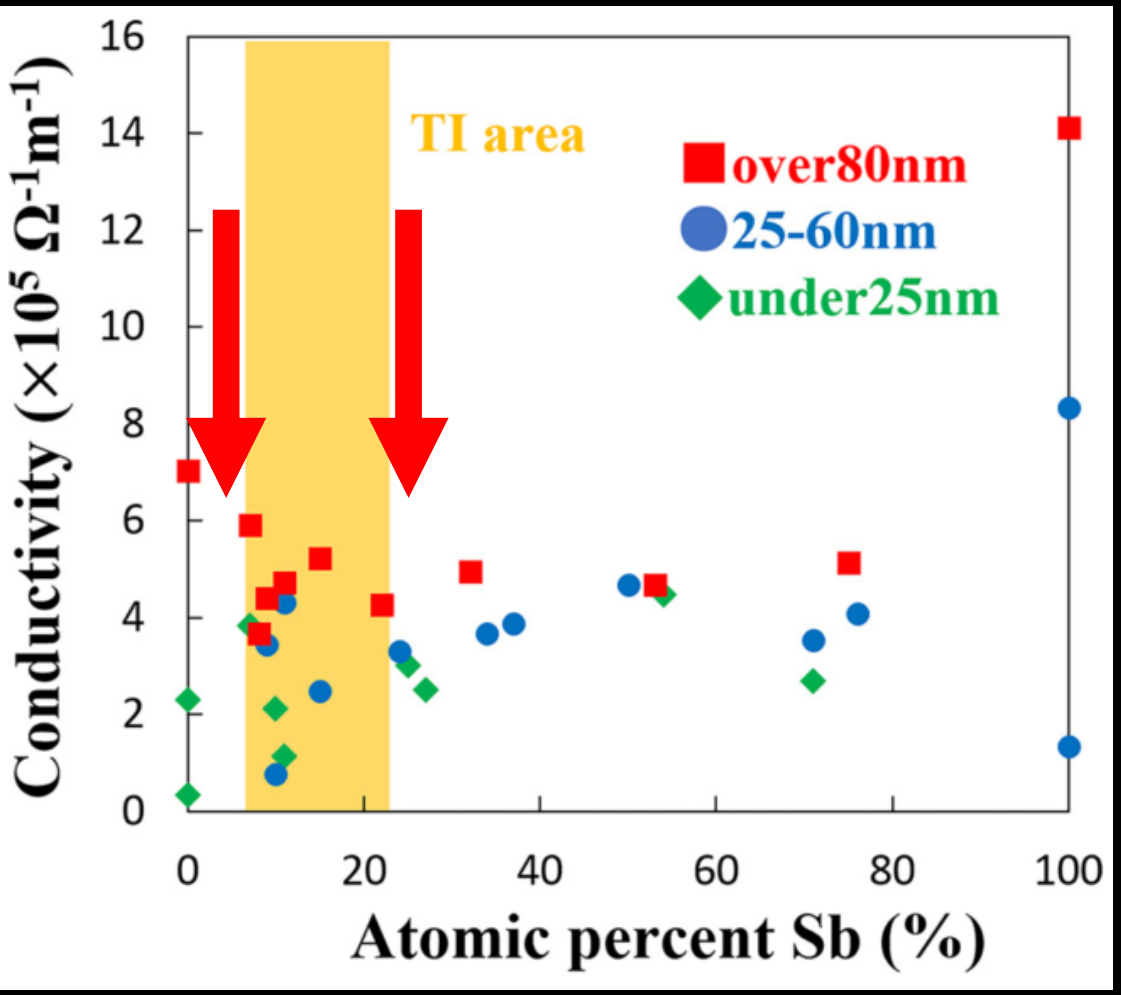
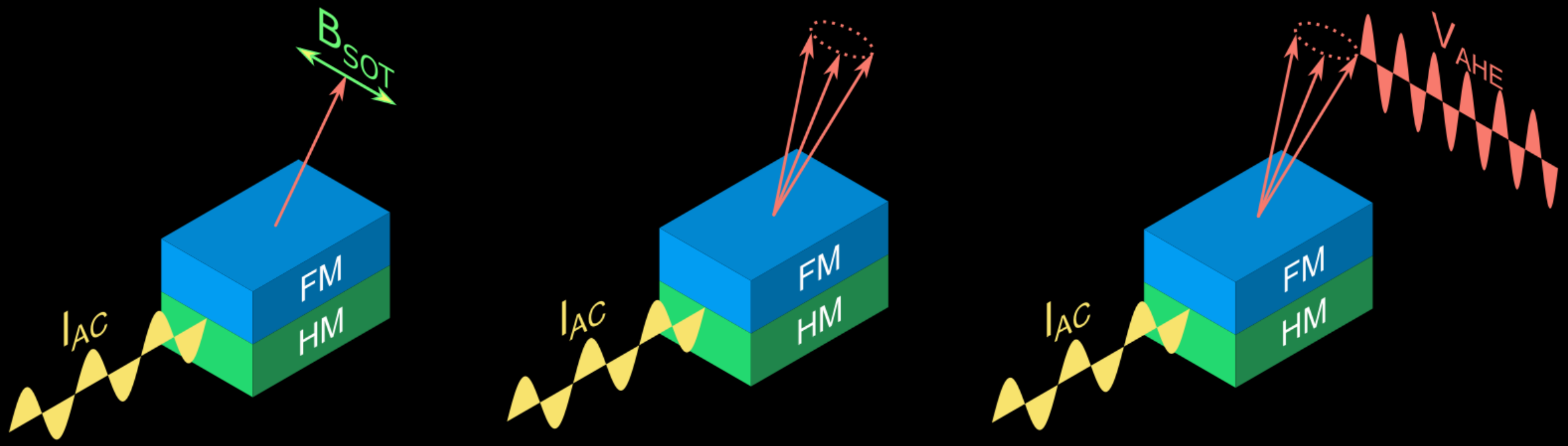
Appendix

Topological properties of Bi-Sb



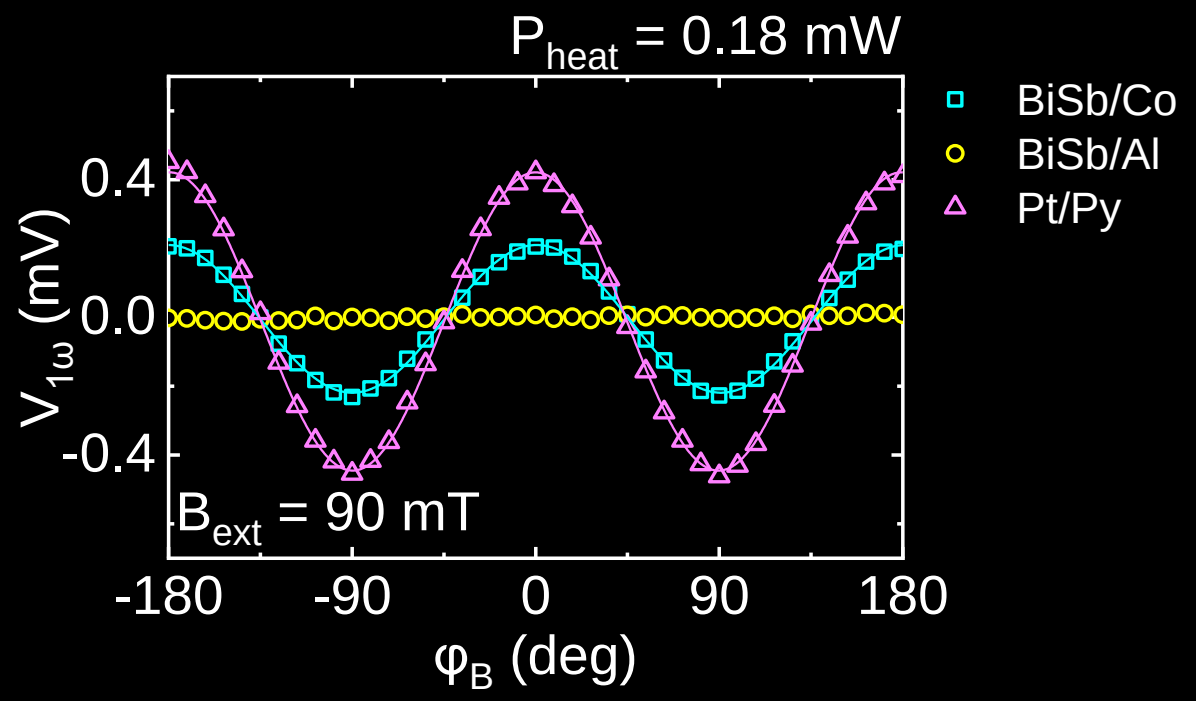
Y. Ueda, N. H. Duy Khang, K. Yao, and P. N. Hai, "Epitaxial growth and characterization of Bi1-xSbx spin Hall thin films on GaAs(111)A substrates," *Appl. Phys. Lett.*, vol. 110, no. 6, p. 62401, Feb. 2017.

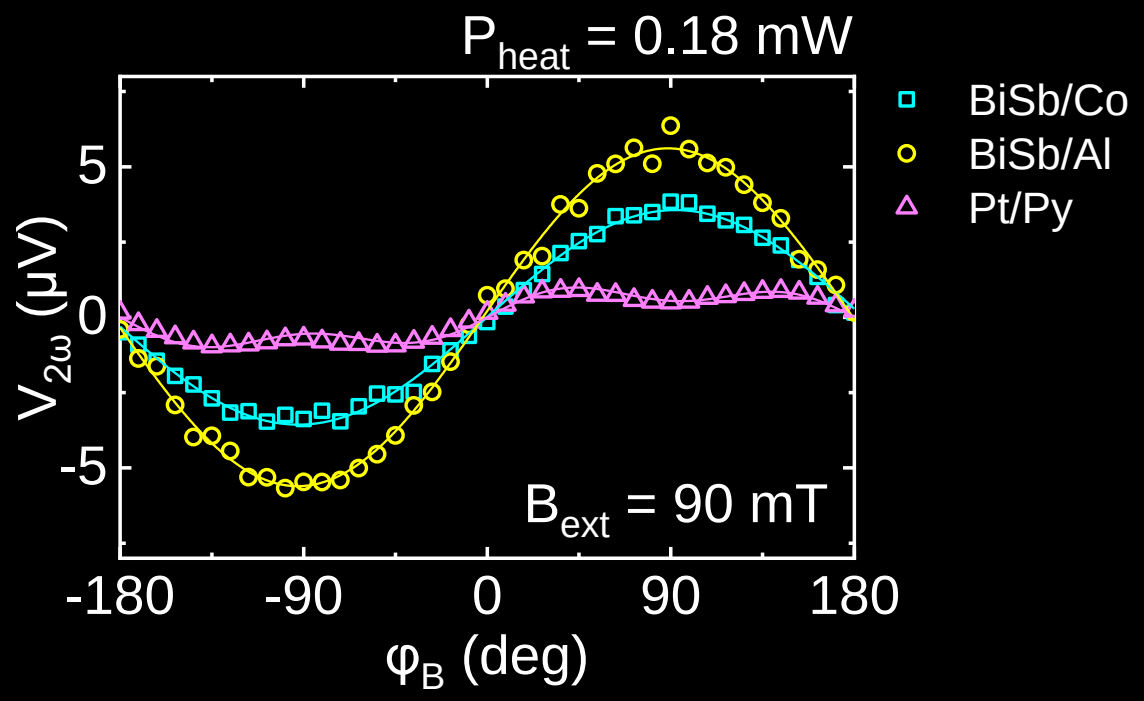
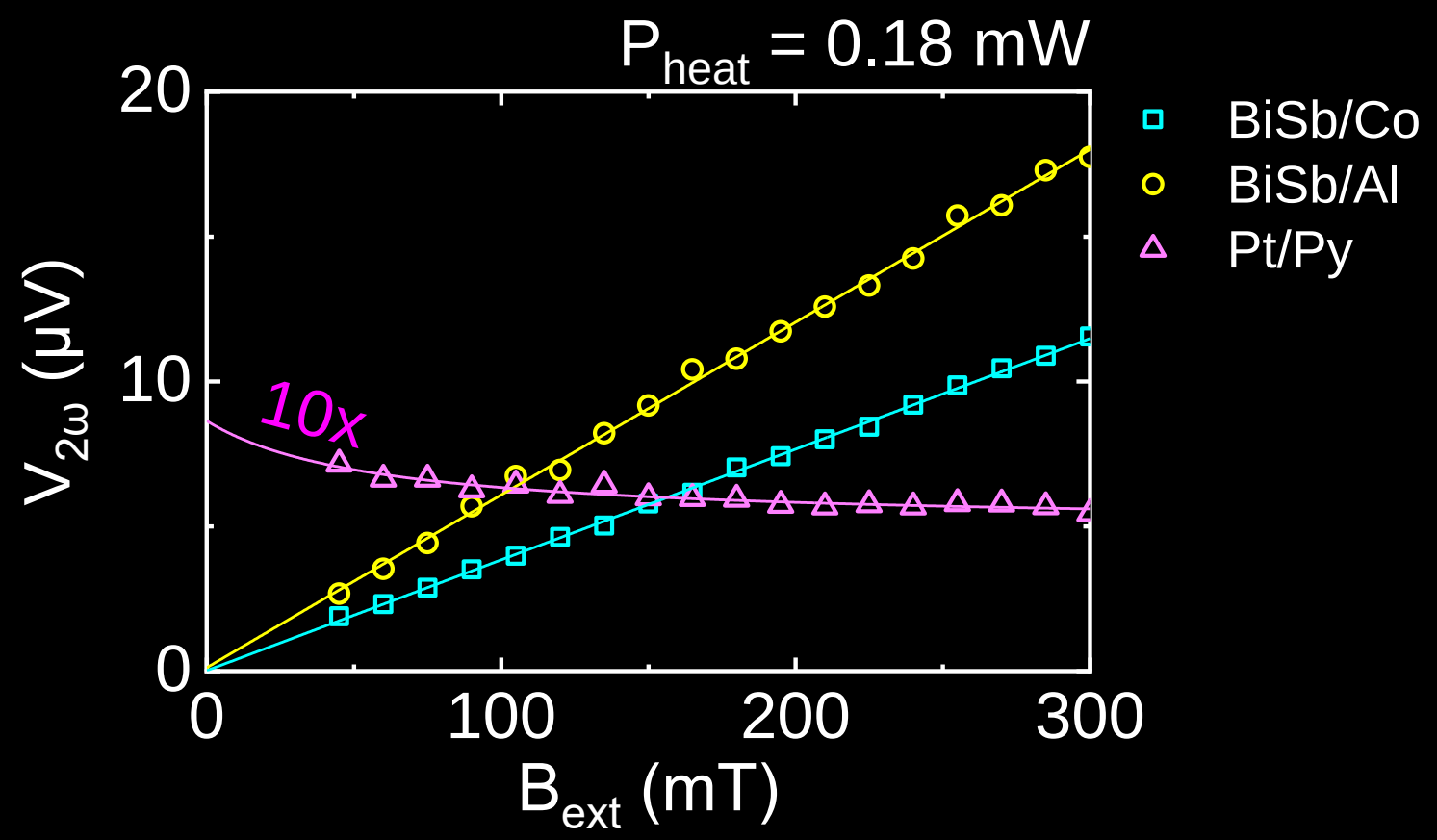
Second Harmonic Hall Voltage



I_{AC} induces B_{SOT} → M oscillates with B_{SOT} → V_{xy} changes periodically

Anisotropic Magnetoresistance and USMR





Anisotropic Magnetoresistance and USMR